COGS 9 Exam 1 Review

TA: Lindsey Gu

Reminders

- **Reading Quiz #3 due next Wednesday**
- **Exam 1: Next Monday @ 2pm, regular classroom**
  - Multiple-Choice questions, 50 mins;
  - The exam will be closed book. Bring a pen/pencil!
- **Group projects**
  - We will sent out your group info later today/tomorrow, make sure to contact group ASAP
  - Checkpoint 1 due **Week5**

Questions?

We have discussed...

- What is data science?
- Ethics and privacy
- Data & data types
- Obtaining data
- Data wrangling

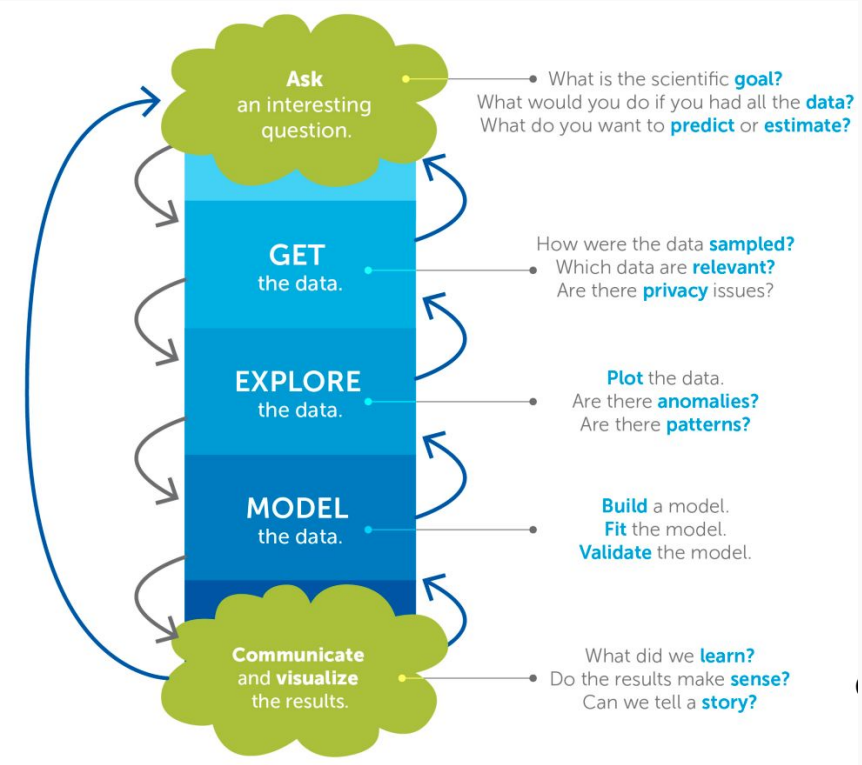
R1: Donoho D, 50 Years of Data Science
R2: Loukides M, Mason, H & Patil DJ, Data's Day of Reckoning
Narayanan and Shmatikov, Myths & Fallacies of "Personally Identifiable Information"
R3: Hadley Wickham, 2014, Tidy Data
Broman & Woo, 2018, Data organization in spreadsheets

Data Science

definition:

The **scientific** and **ethical** process of extracting **value** from **data**

process:



Data Science

Asking a specific, measurable question:

- Topic: Who or what are you studying?
- Outcome or target: What exactly are you measuring?
- Features: What are the factors or comparisons that matter?
- Timeframe: Over what timescale?

Data Science

Goal:

- Describe or understand patterns in the data
- Predict what will happen in the future
- Test cause-and-effect between variables
- Decide what to do or optimize something to get the best outcome

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Code: OMFZUC

Which of the following follows the steps of the Data Science Process in the right order?

(A) Form a hypothesis about movies > scrape IMDB for review data > graph genre vs. sentiment > present results

(B) Collect survey data > ask if age affects pet ownership > plot age against # of cats > train regression model

(C) Neither

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The Target pregnancy prediction case, where a highly accurate model was used to send targeted ads that caused unintended privacy harm, is best described as a failure at which stage of the data science process?

(A) Data collection

(B) Data storage

(C) Data modeling

(D) Deployment

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Donoho, D. “50 Years of Data Science.”

Key Ideas:

- “Identity crisis” and the stats vs. data science debate
  - Big data? Skills? Jobs?
- The “two cultures”
  - Generative, Predictive
- The Common Task Framework (CTF)
  - Netflix Prize
- “Science about data science”
  - cross-study and cross-workflow thinking

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Code: OMFZUC

Which of these would fall under the predictive culture?

(A) An investigation into how social media ads affect voter turnout

(B) Probing a language model to understand its distributions

(C) Building an algorithm to identify music genre

(D) Studying how changes in interest rates have affected unemployment

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Ethics and privacy

Ethics: Principles and standards that guide choices about how data is collected and how models are built and used. What is right, fair, and responsible in a given situation?

Privacy: A person's ability to control access to themselves and their own information. What can others observe, collect, use, or share about you?

Ethics and privacy

The Cobra Effect: the unintended consequences that result from an attempt to solve a problem.

Unintended harms in data science from ...

- data collection: *Should we collect this?*
- data sharing: *Should we share this?*
- data analysis: *Should we analyze this?*
- data modeling: *Should we model or predict this?*

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Code: OMFZUC

Which of these is NOT a good ethical decision?

(A) Spending time to analyze the survey questions when using survey data

(B) Testing your model for equality for different demographics/groups

(C) Monitoring the model for misuse and bias after deployment

(D) Keeping individuals from removing their data after beginning the analysis

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Code: OMFZUC

Which of the following do your devices collect data on (Select all that apply)

(A) Location data (GPS, wi-fi signals, IP address tracking)

(B) Demographic data (age, gender, race)

(C) Search History (Even when on incognito or Private Mode)

(D) Physical interactions (touch patterns, screen time, app usage.

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R2: Data Ethics & Privacy

“Data’s Day of Reckoning”:

- Misuse and abuse of data
- Ethics’ place in the process
- Mechanisms for ethics

“Myths and Fallacies of Personal Identifiable Information”

- The PII Fallacy
- Re-identification
- Provacy Protection

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Code: OMFZUC

(Select all that apply) Which of the following could potentially be PII?

(A) Gender

(B) Amazon purchases

(C) Writing Style

(D) Movie watch history

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Data & data types

Quantitative data:
a value that can be objectively measured

- Continuous (e.g., height, temperature)
  - can have decimals
- Discrete (e.g., number of kids, score in a game)
  - No decimals / fractions

Qualitative data:
describes categories or qualities

- Categorical
- Nominal or unordered sets (e.g., colors, blood type, true/false)
- Ordinal or ordered sets (e.g., low,medium, high)
- Text, audio, images (in raw form)

Data & data types

Quantitative data:

can be visualized as:
- Histogram
- Box plot / violin plot
- Scatter plot
- Line chart (over time)

Qualitative data:

can be visualized as:
- Bar chart
- Grouped/stacked bar chart
- Pie chart (for few categories)

A survey asks students to rate their course satisfaction from 1 (Very Dissatisfied) to 5 (Very Satisfied). What type of variable is this?

(A) Quantitative, continuous

(B) Qualitative, ordinal

(C) Qualitative, nominal

(D) Quantitative, discrete

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A dataset records the ZIP code of each student's home address (e.g., 92122, 10001, 30309). What type of variable is this?

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If a data analyst has data about students' favorite ice cream flavors, which of the following charts is the best option to visualize the data?

(A) Scatter plot
(B) Bar chart
(C) Histogram
(D) Line chart

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Data & data types



Structured data **Semi-structured data** **Unstructured data**

Data & data types

Structured data Semi-structured

can be organized into an n x p table of values

can be stored as:
Spreadsheets: .csv, Excel (.xls, .xlsx)
Databases: SQL (structured query language)

College dataset

$p = 4$
features OR variables

	Private	Apps	Accept	Enroll
Abilene Christian University	Yes	1660	1232	721
Adelphi University	Yes	2186	1924	512
Adrian College	Yes	1428	1097	336
Agnes Scott College	Yes	417	349	137
Alaska Pacific University	Yes	193	146	55
Albertson College	Yes	587	479	158
Albertus Magnus College	Yes	353	340	103
Albion College	Yes	1899	1720	489
Albright College	Yes	1038	839	227
Alderson-Broaddus College	Yes	582	498	172
Alfred University	Yes	1732	1425	472
Allegheny College	Yes	2652	1900	484

qualitative quantitative

Data & data types



Structured data | Semi-structured data | **Unstructured data**

No simple way to represent it in a table format!



Data & data types



Structured data **Semi-structured data** Unstructured data

doesn't fit neatly into rows and columns, but has labels & keys that give it organization

Example:
XML (eXtensible Markup Language)
JSON (JavaScript Object Notation)

XML

```
<empinfo>
   <employees>
      <employee>
         <name>James Kirk</name>
         <age>40></age>
      </employee>
      <employee>
         <name>Jean-Luc Picard</name>
         <age>45</age>
      </employee>
      <employee>
         <name>Wesley Crusher</name>
         <age>27</age>
      </employee>
   </employees>
</empinfo>
```

JSON

```
{  "empinfo" :
   {
      "employees" :  [
      {
         "name" : "James Kirk",
         "age" : 40,
      },
      {
         "name" : "Jean-Luc Picard",
         "age" : 45,
      },
      {
         "name" : "Wesley Crusher",
         "age" : 27,
      }
                  ]
   }
}
```

Obtaining data

1. Collect it yourself
2. Use public / open-source data
3. Buy it
4. Request it from an API
5. Scrape it from a website
6. Source it from your company

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Application Programming Interface
A way for software programs to talk to each other

return *semi-structured data*

Common APIs

- **Maps & locations:** Google Maps
- **Weather:** OpenWeather, WeatherAPI, NOAA (US)
- **Video & music:** YouTube Data API, Spotify Web API
- **Payments:** Stripe API, PayPal API
- **Auth / login:** “Sign in with Google” (Google OAuth)

Obtaining data

1. Collect it yourself
2. Use public / open-source data
3. Buy it
4. Request it from an API
5. **Scrape** it from a website
6. Source it from your company

Web scraping
A way to automatically collect information from websites

Used when there is not API

robots.txt
small text file a website can publish that tells automated crawlers (search engine bots, web scrapers) which parts of the site they're allowed to access

Which of the following is NOT true?

(A) Web scraping never requires privacy consent because the Internet is public

(B) Web scraping is useful when there is no available API

(C) Web scraping must follow website terms of service and privacy rules

(D) Web scraping can automatically collect information from websites

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Which statements about APIs are TRUE? (Select all that apply)

(A) APIs allow software systems to communicate

(B) APIs always provide unstructured data

(C) APIs often return data in formats like JSON

(D) APIs may require keys or have rate limits

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Data wrangling

A significant portion of jobs in data science deal with managing datasets. You should ensure that datasets are tidy, meaning:

- Each variable forms a column
- Each observation forms a row
- Each type of observational unit forms a table

Data wrangling (from lecture)

	A	B	C	D	E
1		marcus j. Pettigew (m)	penny (f)	Jerry Smith	Jessica Chun (f)
2	3/22/2019	2 hours	3		2
3	3/23/2019	3.5 hours	4.6		1.11
4	3/24/2019	3.1 hrs	1	2.2	4.2
5	3/25/2019	**33 hours**	1.5	2	2
6	3/26/2019	0.5 hours	0	5.3	

...

F	G	H	I	J	K	L	M
marcus	penny	jerry	jessica	marcus	penny	jerry	jessica
ios	android	android	ios	good	fine	ok	not bad
ios	android	android	ios	ok	fine	ok	good
ios	android	android	ios	great	fine		good
ios	android	android	ios	great	fine	sad	good
ios	android	android	ios	depressed	fine	happy	meh

Data wrangling (from lecture)

user_information.csv

user_id	name	sex	phone_os
0	marcus j. Pettigew	m	ios
1	penny	f	android
2	Jerry Smith	None	android
3	Jessica Chun	f	ios

user_screen_time_emotion.csv

date	user_id	screen_time_hours	emotion
2019-03-22	0	2	1
2019-03-22	1	3	3
2019-03-22	2	Nan	2
2019-03-22	3	2	5
2019-03-23	0	3.5	5
2019-03-23	1	4.6	3
2019-03-23	2	Nan	None
2019-03-23	3	1.11	3
2019-03-24	0	3.1	1

Good Practices

- Consistency
- Naming conventions
- ISO date/time format
- No empty cells
- Only one piece of information in a cell
- Use plain text (files and formatting)

R3: Tidy Data

“Tidy Data”

- Predictability and reproducability
- Messy data patterns

“Data in Spreadsheets”

- Reliable parsing
- Prevent costly debugging
- Protect raw data integrity

Name	Major, college	Favorite Animal
TQ	DSC, Muir	Cats
Fernando	ECE, Muir	
Gabby	Econ, Warren	Dog
Mia	HDS, ERC	Dogs

In what way is the above table not tidy?
(select all that apply)

(A) Missing values
(B) Multiple variables in one column
(C) Variables in rows *and* columns
(D) Inconsistent values (Dog vs. Dogs)

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